Experiment-11

Implement an application that writes data to the SD card.

Aim:

To develop an Android application that allows the user to write and store data into external storage (SD card) using file handling techniques in Android Studio.

Algorithm:

1. **Start the application**
2. **Create a new Android project**
   - Open Android Studio
   - Select *Empty Activity* and configure project details
3. **Add required permissions**
   - Open `AndroidManifest.xml`
   - Add permission to write to external storage:

     ```
     <uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE"/>
     ```

   - For Android 6.0 and above, request runtime permission in code
4. **Design the user interface**
   - Add:
     - EditText (to enter data)
     - Button (to save data)

1. **Initialize UI components in MainActivity**
   - Link EditText and Button using `findViewById()`
2. **Handle button click event**
   - Set `OnClickListener` on the button
3. **Check external storage availability**
   - Use:
     - `Environment.getExternalStorageState()`
   - Ensure storage is mounted and writable
4. **Create or access file in SD card**
   - Get path using:
     - `getExternalFilesDir()` or `Environment.getExternalStorageDirectory()`
   - Create a file (e.g., `data.txt`)
5. **Write data to file**
   - Use `FileOutputStream`
   - Convert text input into bytes
   - Write bytes into the file
6. **Close the file stream**
   - Ensure proper resource handling using `close()`
7. **Display confirmation message**
   - Show Toast message like “Data saved successfully”
8. **Handle exceptions**
   - Use try-catch blocks for IOException and permission errors

9. **Stop the application**

CODE:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:background="#F5F5F5"
    android:padding="20dp">

    <!-- Card-like container -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="25dp"
        android:background="#FFFFFF"
        android:elevation="8dp"
        android:gravity="center">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="@string/sd_card_storage"
            android:textSize="20sp"
            android:textStyle="bold"
            android:paddingBottom="20dp"/>

        <EditText
            android:id="@+id/editText"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:autofillHints=""
            android:background="@android:drawable/edit_text"
            android:hint="@string/enter_text_to_save"
            android:inputType="text"
            android:padding="12dp"
            android:textColorHint="#757575"
            tools:ignore="VisualLintTextFieldSize" />

        <Button
            android:id="@+id/btnSave"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="@string/save_to_storage"
            android:layout_marginTop="20dp"
```

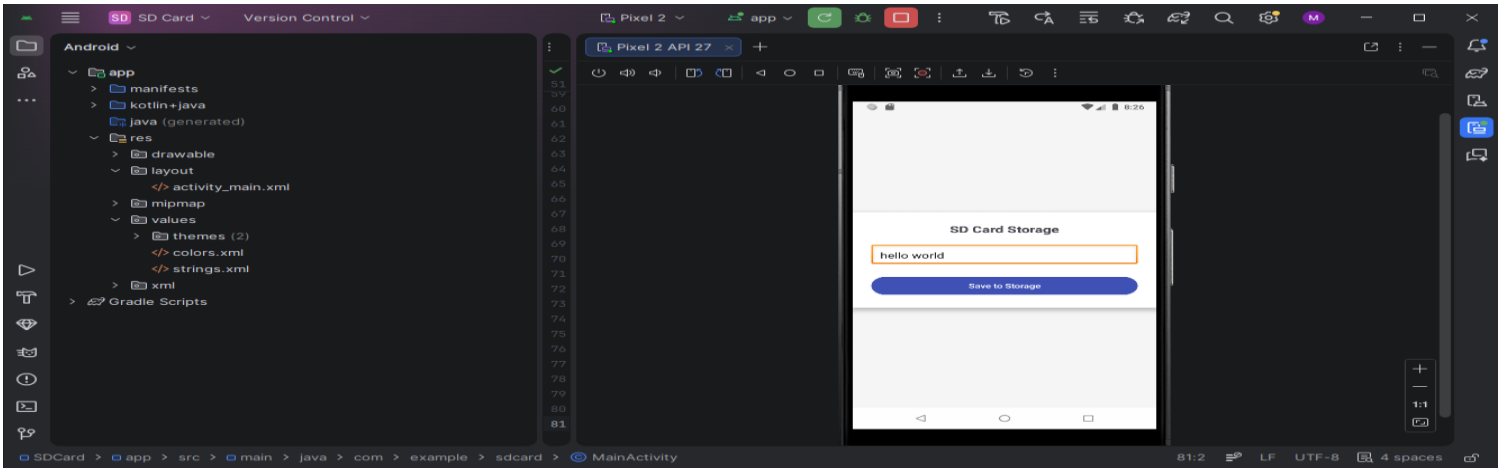
```
            android:backgroundTint="#3F51B5"
            android:textColor="#FFFFFF"
            tools:ignore="VisualLintButtonSize" />

    </LinearLayout>

</LinearLayout>
```

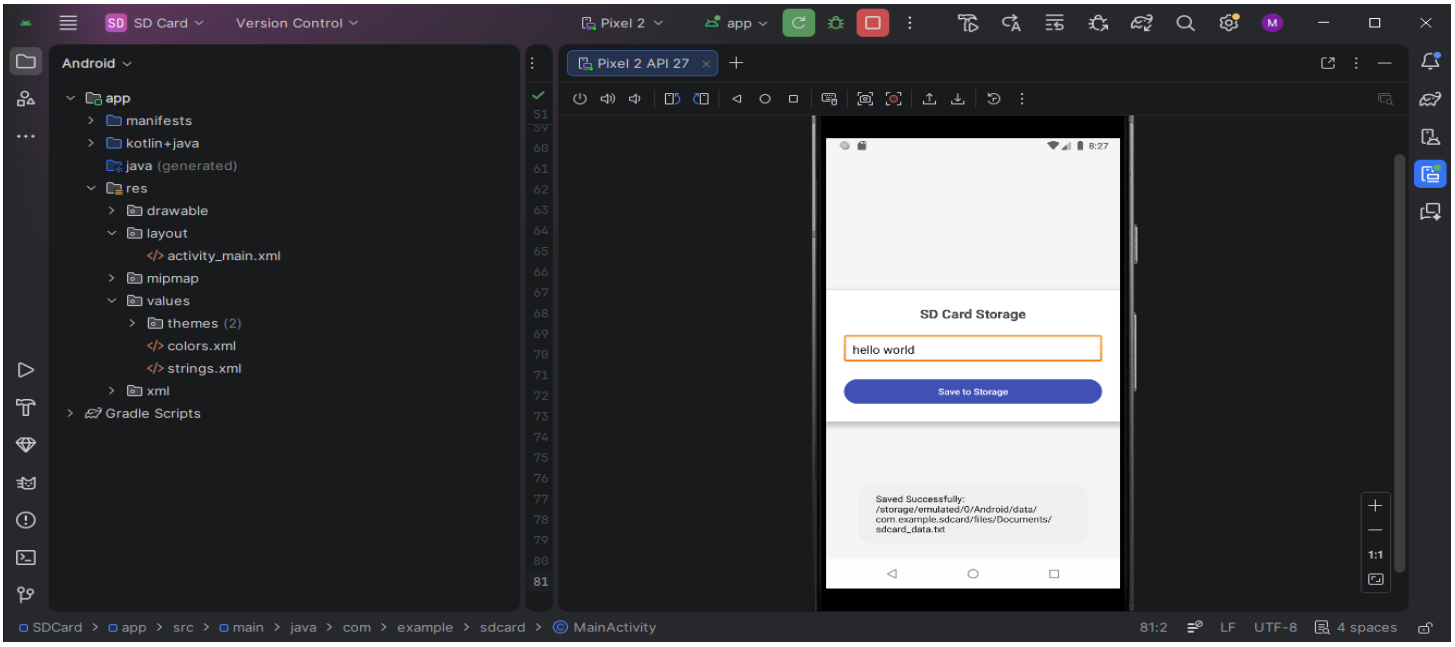
MainActivity.java:

```
package com.example.sdcard;

import android.os.Bundle;
import android.os.Environment;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import java.io.File;
import java.io.FileOutputStream;

public class MainActivity extends AppCompatActivity {

    EditText editText;
    Button btnSave;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        // Edge-to-edge padding
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });

        editText = findViewById(R.id.editText);
        btnSave = findViewById(R.id.btnSave);

        btnSave.setOnClickListener(v -> {
            String data = editText.getText().toString().trim();

            if (data.isEmpty()) {
                Toast.makeText(this, "Please enter text", Toast.LENGTH_SHORT).show();
                return;
            }
```

```java
            saveToStorage(data);
        });
    }

    private void saveToStorage(String data) {
        try {
            // Modern safe storage (recommended)
            File folder = getExternalFilesDir(Environment.DIRECTORY_DOCUMENTS);

            if (folder != null && !folder.exists()) {
                boolean created = folder.mkdirs();

                if (!created) {
                    Toast.makeText(this, "Folder creation failed", Toast.LENGTH_SHORT).show();
                    return;
                }
            }

            File file = new File(folder, "sdcard_data.txt");

            FileOutputStream fos = new FileOutputStream(file);
            fos.write(data.getBytes());
            fos.close();

            Toast.makeText(this,
                    "Saved Successfully:\n" + file.getAbsolutePath(),
                    Toast.LENGTH_LONG).show();

        } catch (Exception e) {
            Toast.makeText(this,
                    "Error: " + e.getMessage(),
                    Toast.LENGTH_LONG).show();
        }
    }
}
```

OUTPUT:





Result:

The Android application was successfully developed using Android Studio to write data to the SD card (external storage).